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Battery Energy Storage System: A Milestone in Mumbai's Grid Evolution

Presented By

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INTRODUCTION

Energy Storage Systems (ESS) will be key to balance the system and deliver firm power as India rapidly progresses to high levels of Variable RE (VRE) in total electricity generation

% of VRE of total electricity supply by Phase Transition		Grid Integration Challenge	Solution/Approach
50%+	Growing amounts of seasonal VRE surplus	Increased risk of substantial VRE curtailment	*Adding more VRE might not be feasible, hydrogen, nuclear may be better solutions*
25 to 50%	System experiences periods with high VRE dependency	Greater swings in the supply-demand balance	*Changes to regulatory & operational approach to integrate large amounts of ESS for RTC power*
10 to 25%	VRE determines operation pattern of the system	Differences between load and net load noticeable and changes in power flow patterns	Upgrades to operating practices, better use of existing system flexibility, ESS integration needed strategically across value chain
5 to 10%	VRE has moderate impact on the system		
<1%	VRE has no noticeable Impact on the system	Manageable changes to the system at a localised level	

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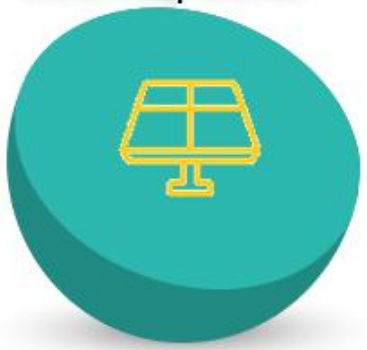
India by 2030

Central Electricity Authority of India has suggested deployment of substantial ESS by 2030 to integrate 500 GW of renewables

NEED FOR BESS

Renewable Energy Integration – Grid & Rooftop Solar



Power purchase cost optimization

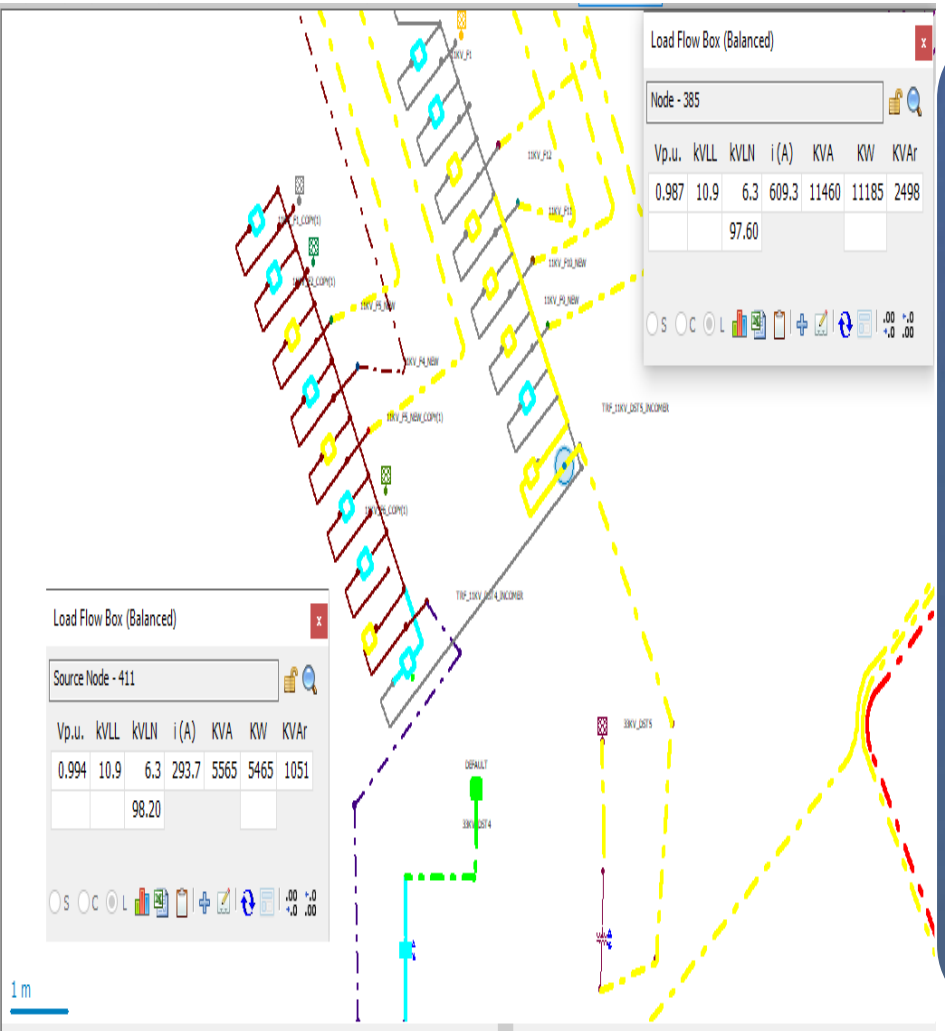
Peak Load Management

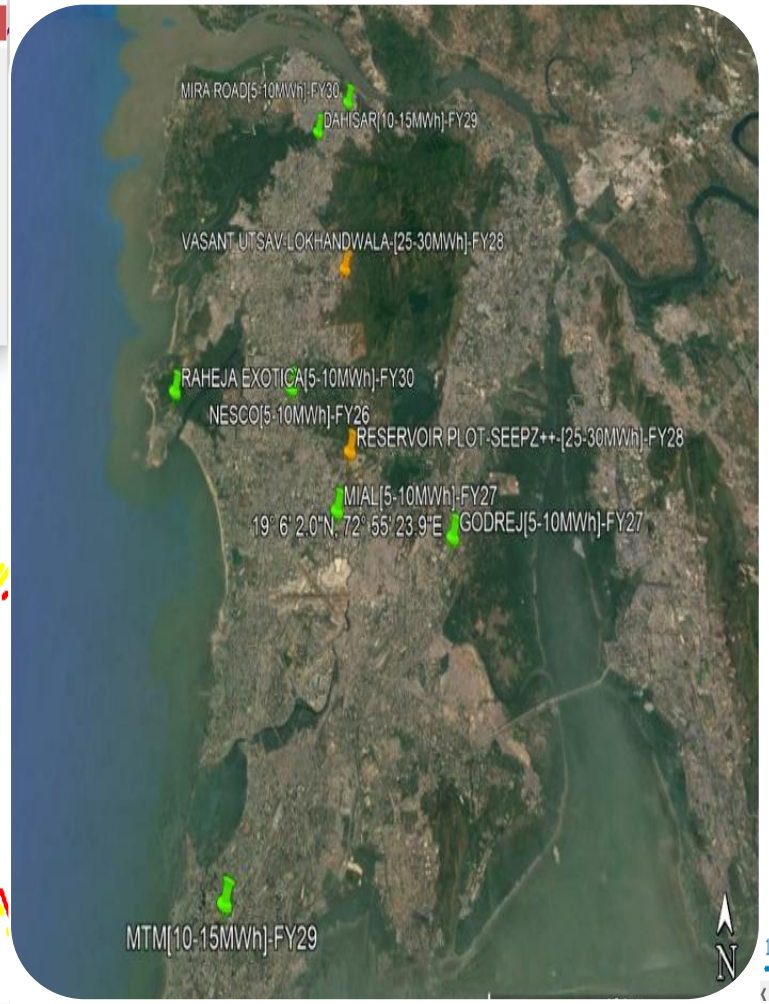
Green Energy

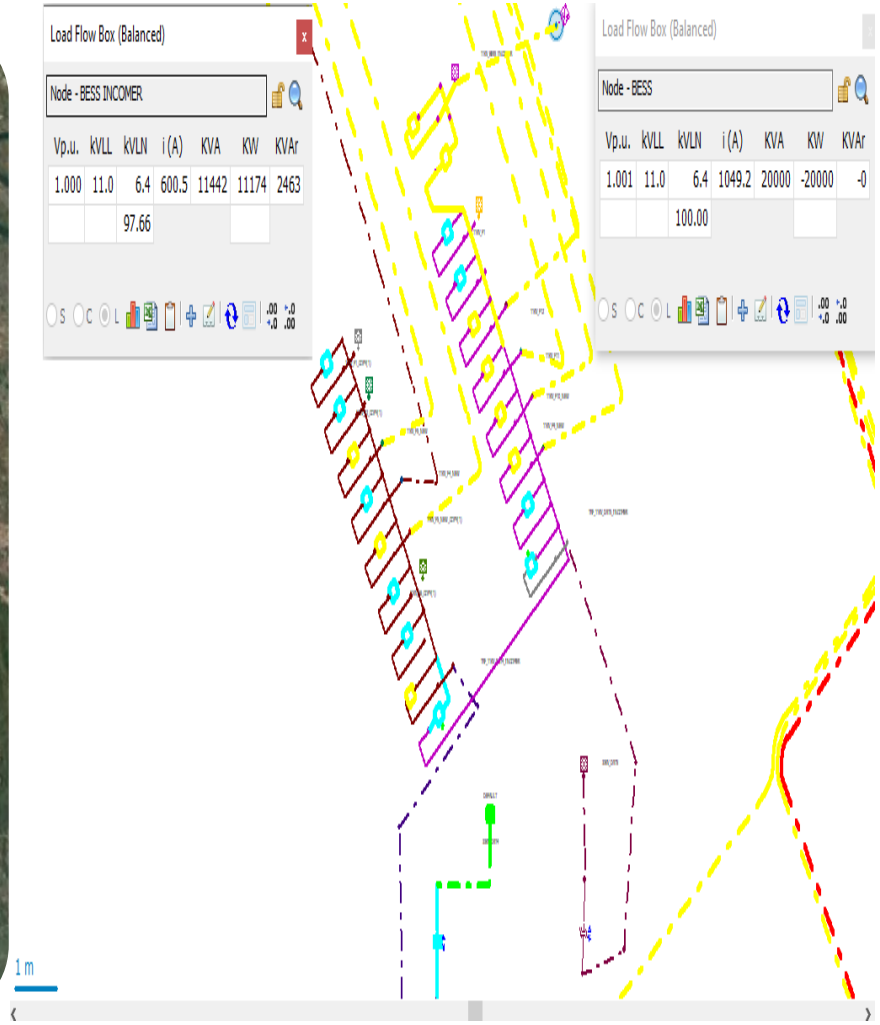
Opening of ancillary market in India(RRAS)

Reliability &Network Congestion

FEASIBILITY STUDY FOR STRATEGIC DEPLOYMENT



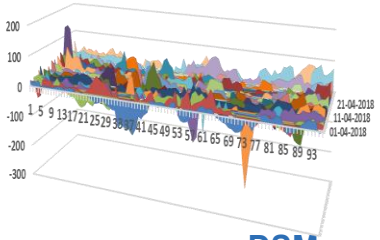




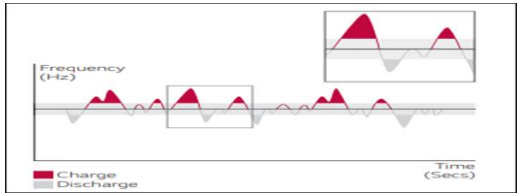
BESS GRID APPLICATIONS

Capex Deferral

Black Start

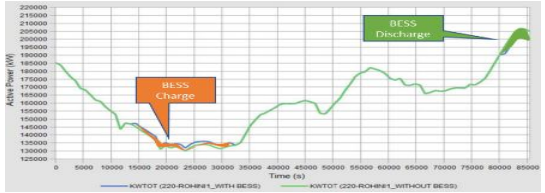


DSM

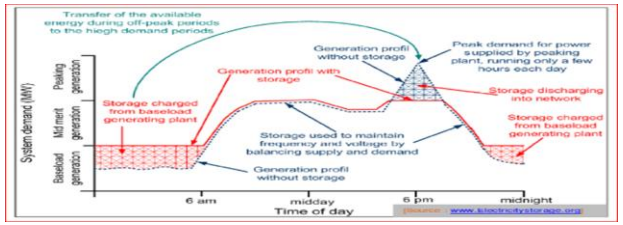


Frequency Regulation

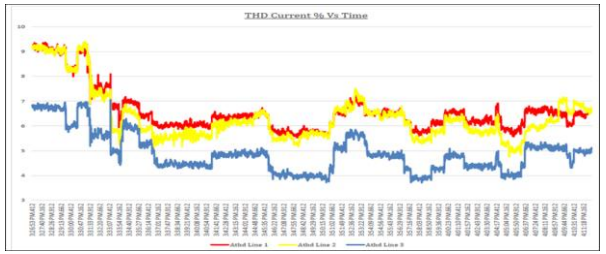
BESS



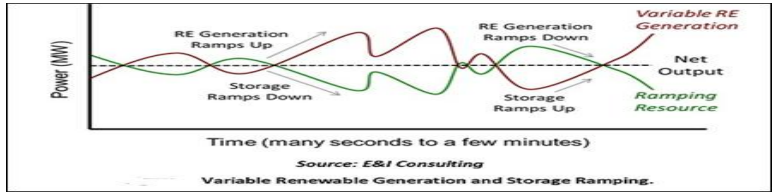
Energy Arbitrage



Peak Load Management



Power Quality/Reactive Power

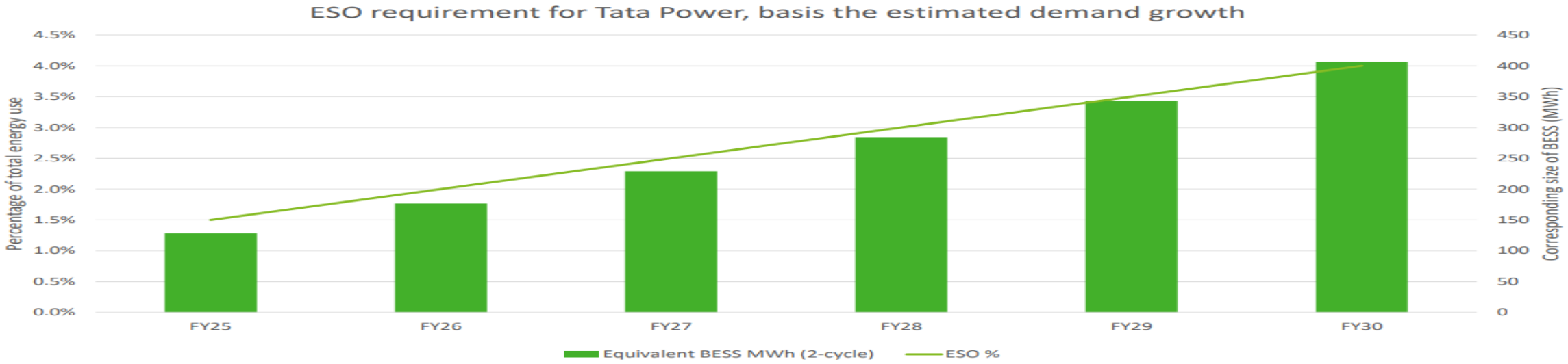


Renewable Integration

ESO OBLIGATION

Energy Storage Obligation (ESO) has been mandated for all DISCOMs in Maharashtra from FY25

Energy Storage Systems must also be charged with sufficient renewables in order to meet the compliance



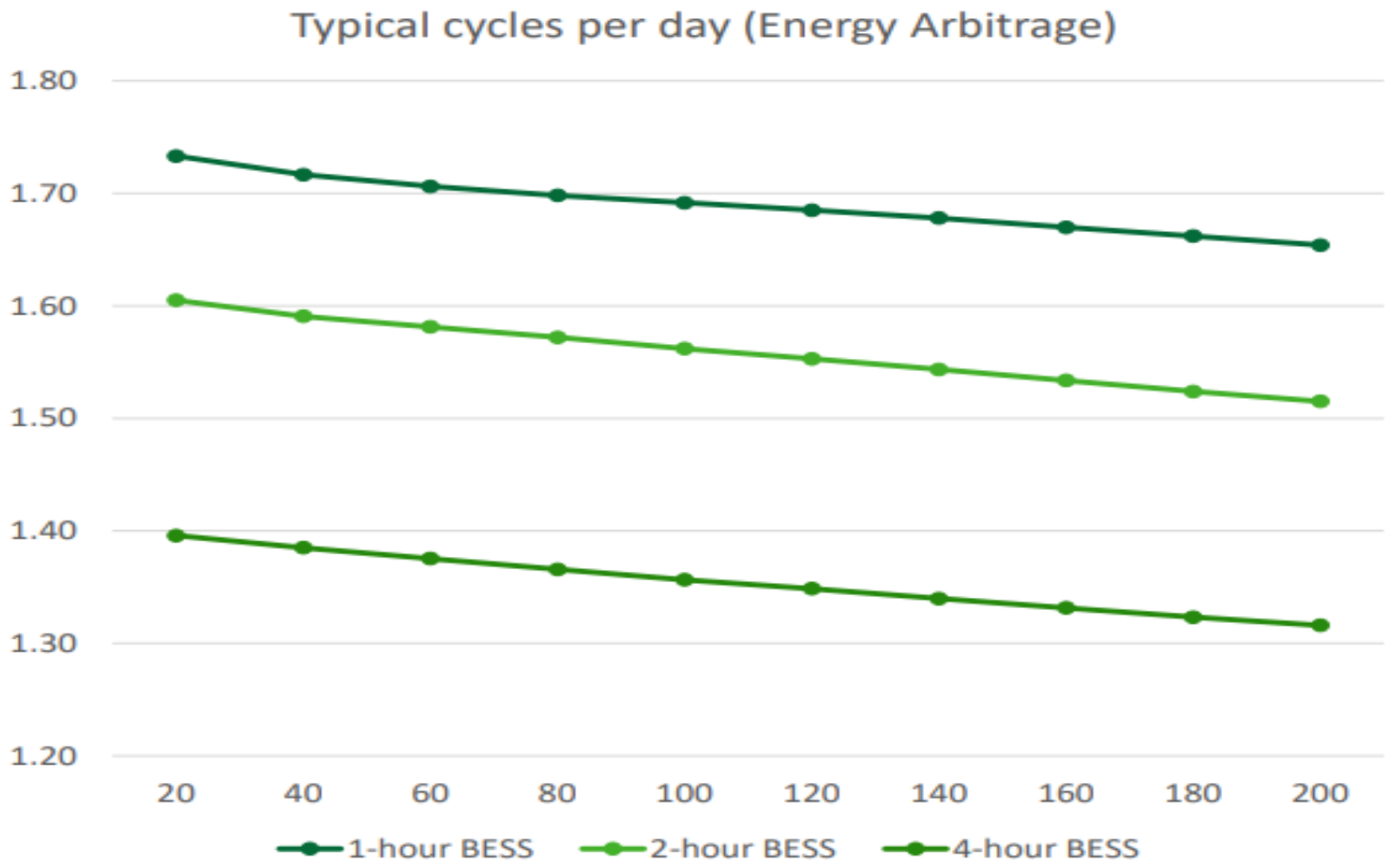
Renewable Purchase Obligation (RPO)

Year	Quantum of purchase (in %) from Renewable Energy sources (in terms of energy equivalent in kWh)				
	Wind RPO	HPO	Distributed RPO	Other RPO	Total RPO
	(a)	(b)	(c)	(d)	(e)
2024-25	0.67%	0.38%	1.50%	27.35%	29.91%
2025-26	1.45%	1.22%	2.10%	28.24%	33.01%
2026-27	1.97%	1.34%	2.70%	29.94%	35.95%
2027-28	2.45%	1.42%	3.30%	31.64%	38.81%
2028-29	2.95%	1.42%	3.90%	33.10%	41.36%
2029-30	3.48%	1.33%	4.50%	34.02%	43.33%

Energy Storage Obligation (ESO)

Year	Storage (on Energy basis)
2024-25	1.5%
2025-26	2.0%
2026-27	2.5%
2027-28	3.0%
2028-29	3.5%
2029-30	4.0%

BESS & ENERGY ARBITRAGE



- More flexibility in cycle with Distributed capacity
- No of cycle: 500 cycles/year

- Key benefits: Reduction in Power purchase cost, Renewable curtailment and Reduced tariff for consumers

BESS & TRAS

Use of BESS for tertiary reserve ancillary services

Commitment of BESS to TRAS markets can provide additional revenue compared to arbitrage only

	TRAS – UP (discharging mode)	TRAS – DOWN (charging mode)
When is it utilized?	Low grid frequency (<49.90 Hz)	High grid frequency (>50.05 Hz)
Reason for frequency deviation	Supply < Demand	Supply > Demand
Corrective measure	*"**increase active power injection or decrease drawal or consumption**, as the case may be, in response to despatch instructions of the Nodal Agency"*	*"**reduce active power injection or increase drawal or consumption**, as the case may be, in response to despatch instructions of the Nodal Agency"*
Eligibility – BESS Idle hours	Yes – As long as sufficient SOC is available (greater than 0)	Yes – As long as sufficient SOC is available (Less than max)
Eligibility – BESS Charging hours	No	Maybe (if BESS discharge rate is not already at maximum)
Eligibility – BESS Discharging hours	Maybe (if BESS charging rate is not already at maximum)	No
Share of committed hours dispatched	**7%** (based on previous years' data from POSOCO)	**15%** (based on previous years' data from POSOCO)
Commitment charge	**Rs. 0.20/ kWh**	N/A

KEY OUTCOMES

- **Energy arbitrage and ramping support** is the most lucrative value stream, since peak hour market purchases (near Rs. 10/ kWh cap) can be reduced in favour of BESS charged with cheaper renewables
- The BESS can also provide **ancillary services** by commitment to TRAS markets during hours when it is not utilized for Energy Arbitrage
- BESS can help with **capex deferral** at the distribution substations by catering to peak demand and deferring transformer augmentation
- **Islanding through BESS** can provide reliable power to critical loads near substations during outages, however, details from the load centers will be required to quantify the benefits

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THANK YOU

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